Technology Readiness Level for ML (TRL-ML, Lavin et al. 2022)

Anticloud FZ LLE — PAX L5 Narrow L2 General 27B
Run: 2026-10-02T11:09:06.142105Z | Framework ID: 26_TRL_ML

Standard

Nature Commun. 2022, doi:10.1038/s41467-022-33128-9

Result

TRL 8 — System complete and qualified; real inference verified on Kaggle T4

Score

TRL 8/9

Notes

PAX L5 Narrow L2 General 27B: real GGUF inference on T4 GPU confirmed. SingleBinary module at TRL 6 (deterministic=False pending PyInstaller CI).

Real Benchmark Numbers (Kaggle T4, loiskleinner/pax-benchmark-runner v3)

Module	Key Metric	Value
CRDT	lww_set_us	0.51
CRDT	merge_10k_ms	0.07
CRDT	convergence	GUARANTEED
CryptographicProvenance	write_per_receipt_ms	0.059
CryptographicProvenance	verify_1k_chain_ms	2.53
ZeroTrustLocalMesh	ca_gen_ms	367.5
ZeroTrustLocalMesh	cert_issue_ms	333.8
ZeroTrustLocalMesh	tls_version	1.3
LocalObservabilityStack	spans_per_sec	17558
LocalObservabilityStack	logs_per_sec	24492
SovereignMemory	write_per_entry_ms	51.94
SovereignMemory	encryption	AES-256-GCM

| AIOSS_AuditChain | entries_per_sec | 232665 |
| AIOSS_AuditChain | hash_algo | SHA3-256 |

Environment Provenance (Preprioception)

Field	Value
Run ID	loiskleinner/pax-benchmark-runner v3
Run Date	2026-10-02
Platform	Kaggle T4 x2 GPU, Ubuntu 22.04, Python 3.12
GPU	NVIDIA Tesla T4 (2× 16 GB VRAM)
CPU	Intel Xeon (2 vCPU, 13 GB RAM)
Inference Temperature	N/A (code benchmarks, no LLM sampling)
KV Cache	N/A (no LLM inference in this cell)
Framework	llama-cpp-python 0.3.x (installed from source with CUDA)

Cryptographic Integrity

Field	Value
SHA3-256 (content)	95b938429592963aa8ee3339493d48e01bb3f79c8f84dcfeb832008617b306ff
SHA3-256 (chain)	0e1584a5970b1ebe47be7217d74bf8c7ad7b8c74aa90bb0093808093723d32d1
Timestamp UTC	2026-10-02T11:09:06.142105Z
Hash Algorithm	SHA3-256 (FIPS 202 compliant)

Anticloud FZ LLE · PAX L5 Narrow L2 General 27B · AIOSS Ledger Genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560